

COMP8420 2026 S1

Assignment 2: Large Language Models

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Due: Friday 17 April 2026



This assignment assesses the following unit learning outcomes:

- ULO1: Compare and evaluate the key natural language processing applications that meet the current and emerging industry needs.
- ULO3: Implement natural language processing applications using common tools and libraries used in industry.
- ULO5: Apply natural language processing methods and techniques to industry applications using real data.

**It is expected that students will need approximately 30 to 40 hours to complete the project *individually*.
It accounts for 30% of the overall assessment for this unit.**

Start this assignment as soon as possible! The lectures and workshops will enable you with tech and hints for undertaking the assignment. You would be expected to plan and make weekly progress and discuss with our tutors in the workshops.

Essential Timeline:

- Release Date: Week 2, Friday 6 March 2026
- Due Date: Friday 17 April 2026 (during the Easter holidays)

1 Introduction

Assignment 2: Large Language Modeling invites you to create a smart personal financial (or other) assistant that offers customized financial suggestions based on your client's preferences and requirements. Your client aims to build an application powered by large language models (LLMs) that generates personalized financial strategies, trading rules, planning, or recommendations, etc. Your responsibilities include:

1. Setting Up a Language Model: Select, configure, finetune and integrate a suitable (pretrained) language model for processing finance-related queries.
2. Automating Dialogue: Develop an automated dialogue system using LLM techniques to interact and cope with end users' financial requirements through LLM.
3. Developing a Frontend: Create a user-friendly interface to engage users, and visualize and manage the financial assistance requirements and processes.
4. Demonstrating LLM Effectiveness: Showcase how your customized LLM benefits the client with advanced system performance and impact in your report.

2 LLM Setup

For this project, you are required to configure a large language model (LLM) as the backend. You will deploy a local pretrained LLM model. If online agentic design is incorporated, ensure that all connections are properly established and demonstrate the techniques and functions of your LLMs by showing various sample prompts and results for financial assistance.

Rubric [maximum 10 marks]:

- Select, justify and compare your preferred pretrained LLMs [2 marks]
- Configure, finetune, and deploy local LLMs in your program context [4 marks]
- Collect financial assistance related data, showcase your LLM techniques [3 marks]
- Showcase functions and performance of your LLMs (e.g. prompting) [1 mark]

Note: Deductions may apply if errors occur during the LLM deployment, data collection, etc. processes.

3 LLM-driven Personal Assistant Dialogue System

The primary objective is to develop a smart LLM-enabled personal assistant that

- helps users understand financial assistance (e.g. planning, strategies) task;
- collects essential data and delivers insightful strategies, recommendations.

To achieve this, you are required to develop an LLM-driven dialogue procedure that can:

- collect client's personal details, and financial requirements and preferences;
- process the gathered data to produce tailored financial strategies, recommendations based on your client's request using the LLMs;
- improve interaction and financial assistance quality by comparing different LLM models, experimenting with different LLM techniques (e.g. prompting, CoT), and incorporating client and external financial prices, announcements, news, etc.

Rubric [maximum 20 marks]:

- Demonstrate a working dialogue with the LLMs [2 marks];
- Showcase a friendly frontend [3 marks], where Python Notebook is worth 1 mark and an advanced visualization (website or mobile app) is worth 3 marks;
- Develop a friendly LLM dialogue chain that collects key information [8 marks]: task definition [2 marks], personal information [2 marks], financial preferences [2 marks], and additional LLM techniques to enhance performance or user experience [2 marks];
- Present a well-structured and justified financial recommendation as the output [2 marks];
- Investigate different settings by comparing various LLMs and techniques [3 marks];
- Justified results, a professional report, and a video [2 marks].

Note 1: Deductions will apply if errors occur during the interaction, response, analysis and recommendation process.

Note 2: Given its high impact on your overall score, allocate sufficient time and effort to this task.

Note 3: A report showing your LLM development, processes, results and core codes is required. If you create a Python notebook, it can serve as supporting your report. It is expected that a mobile app or a website would be created, enabled by your LLMs. Your final report should include the entire system design, implementation, demonstrations, interfaces, results, and source codes.

4 Advancing Your System

Higher marks for this project will be attributed to various advancements in your system, including but not limited to excellence in the following areas:

- Compare LLMs with justification, demonstrating advanced LLM techniques.
- Showcase the ability to support multiple languages.
- Propose innovative LLM techniques with justified performance for financial assistance.
- Accommodate diverse user requirements for financial assistance.

Note: The total assignment score is capped at 30 points.

5 Submission

You are required to submit a comprehensive project, the report, and a video to iLearn for marking. The project should contain your solution (with code, data, results, etc.) and the report. The report should be in '.pdf' with source codes (e.g. '.ipynb', '.py') files. The video should record a full demonstration of your working system within (maximally) 5 minutes. Please zip the report, video and codes under 5 MB (if possible) with the following file organization structure.

Your submission may be organized as follows:

```
report/StudentID_Report.pdf
report/video
codes/
  Notebook1.ipynb
  Notebook2.ipynb...
```

6 Notes

Should I use PyTorch or TensorFlow? Both frameworks are acceptable for our assignments. Please choose the one with which you are most comfortable. In alignment with our workshop settings, we highly encourage you to get familiar with PyTorch as soon as possible.

Can I use other tools in my assignments? In general, any publicly available Python package is permitted. However, you must clearly specify the tools and libraries you utilize, finetune, and adjust for your assignment.

Can I reuse previous projects or assignments from other units?

No, you are NOT allowed to reuse code pieces from other units. In this LLM assignment, you are expected to select, tune, and apply your own pretrained LLM, and develop a system for the financial assistance purpose.

Can I use materials from the workshops? Absolutely. You are encouraged to apply, modify, and implement any tools introduced during our workshops. Keep in mind, though, that simply copying and pasting code will not solve the expected purpose. You would adjust them for this assignment. Simple applications will significantly influence your final score.

What is the collaboration policy? This assignment must be completed individually, and each student is required to submit their own work via iLearn. While discussing ideas with classmates is permitted, your final submission must be entirely your own work.

7 Policy

Late Submission Policy

For both assignments and the major project, NO late submission policy applies, unless Special Consideration has been submitted and approved. All deadlines mean 11:59:59 pm that day.

- **5% penalty per day:** If you submit your assessment late, 5% of the total possible marks will be deducted for each day (including weekends), up to 7 days.
 - *Example 1 (out of 100):* If you score 85/100 but submit 20 hours late, you will lose 5 marks and receive 80/100.
 - *Example 2 (out of 30):* If you score 27/30 but submit 1 day late, you will lose 1.5 marks and receive 25.5/30.
- **After 7 days:** Submissions more than 7 days late will receive a mark of 0.
- **Extensions:**
 - Automatic short extension: Some assessments are eligible for automatic short extension. If so, you can only apply for an automatic short extension before the due date.
 - Special Consideration: If you need more time due to serious issues and for any assessments that are not eligible for Short Extension, you must apply for Special Consideration.

Need help? Review the Special Consideration page [HERE](#).

Special Consideration Policy: The Special Consideration Policy aims to support students who have been impacted by short-term circumstances or events that are serious, unavoidable and significantly disruptive, and which may affect their performance in assessment. If you experience circumstances or events that affect your ability to complete the assessments in this unit on time, please inform the convenor and submit a Special Consideration request through ask.mq.edu.au.